SubRotQ: KV Cache Compression via PCA Projection and Random Rotation Quantization

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Abstract

We study compression of the key-value (KV) cache in transformer language models via projection into a learned PCA subspace followed by random rotation and uniform quantization (SubRotQ). Through 30 experiments across five architectures (Qwen3-1.7B/14B/32B, Mistral-7B, Phi-4, Llama-3.1-8B), we establish that **truncation error from dimension reduction categorically dominates quantization noise**: on WikiText-2, k=64, 16-bit (pure truncation) produces $6.25\times$ perplexity degradation while k=128, 4-bit (no truncation, aggressive quantization) produces only $0.98\times$. This motivates a compression target of k=128 (full rank, quantization only), yielding $4.00\times$ **compression at** $< 2\%$ **perplexity cost** on Qwen3-14B. We further find that **V compression is not viable at any** $k < d_{\text{head}}$ **across all tested architectures**—including Llama-3.1 which lacks QK-norm—indicating that V vectors are intrinsically high-dimensional, not an artifact of any specific normalization scheme. K compression generalizes across architectures and context lengths ($4\times$ compression, $<2\%$ PPL on Qwen3/Mistral/Llama; stable to 40K tokens; near-lossless on downstream reasoning). We recommend **K-only compression at** k=128**/4-bit** as a production-ready configuration.

1 Introduction

Inference with large language models is increasingly memory-bound at long context lengths. For a 14-billion-parameter model like Qwen3-14B at 32K tokens, the KV cache alone requires approximately 10 GB of GPU memory—comparable to the model weights themselves. This limits the practical context window on consumer hardware and increases serving costs proportionally with sequence length.

KV cache compression aims to reduce this memory footprint at inference time without modifying model weights. The central challenge is that KV vectors must be reconstructed with sufficient fidelity to preserve attention score distributions, which determines the model's output quality.

Our starting point is the **Universal Weight Subspace Hypothesis (UWSH)** Kaushik et al. (2025), which shows that deep networks trained on diverse tasks converge to shared low-dimensional parametric subspaces. We ask whether this structure extends to KV *activation* vectors at inference time: if KV vectors occupy a stable, low-dimensional subspace per (layer, head), offline PCA calibration should suffice for compression. Our experiments confirm this—K vectors achieve cross-domain overlap of 0.70 and cross-layer overlap of 0.56, supporting the viability of offline subspace compression.

We combine subspace projection with **random rotation quantization** (SubRotQ): a random orthogonal matrix R is applied before uniform scalar quantization to spread variance across dimensions and reduce quantization error. This is distinct from PolarQuant Han et al. (2025), which uses recursive polar coordinate transformation with k-means codebooks.

Our primary contribution is an empirical characterization of this method across scales and architectures, producing three actionable findings:

1. **Truncation error is the binding constraint.** The critical design variable is k (subspace dimension), not bit depth. At k=128 (full rank), 4-bit quantization achieves $< 2\%$ WikiText-2 PPL degradation. Practitioners should maximize k before minimizing bits; reducing bit depth compensates far less than restoring dimensions.

2. **V compression is universally intractable.** Key vectors compress well across all tested architectures. Value vectors resist subspace compression at any $k < d_{\text{head}}$, regardless of whether the architecture uses QK-norm. This is an intrinsic property of V variance structure.

3. **K-only compression at k=128/4-bit is the production configuration.** Full-rank 4-bit quantization of K achieves $4.00\times$ compression with $< 2\%$ WikiText-2 PPL degradation, generalizes across Qwen3, Mistral, and Llama, remains stable through 40K-token contexts, and produces near-lossless downstream task accuracy.

2 Background and Related Work

KV cache compression. Prior work falls broadly into four categories. *Token eviction* Zhang et al. (2023); Xiao et al. (2024) discards KV entries for less important tokens. *Quantization* Hooper et al. (2024); Liu et al. (2024); Kang et al. (2024) reduces bit precision of stored vectors. *Dimensionality reduction* projects KV vectors to lower-rank representations: KVTC Staniszewski and Łańcucki (2026) combines PCA with dynamic-programming bit allocation and entropy coding for $20\times$ compression; Palu Chang et al. (2025) decomposes KV projection matrices via SVD; MatryoshkaKV Park et al. (2025) trains orthogonal projections via knowledge distillation; Eigen Attention Yang et al. (2024) performs attention directly in PCA-reduced space; SVDq Wu et al. (2025b) uses SVD with importance-aware mixed precision; SQuat Li et al. (2025) combines subspace projection with orthogonal rotation quantization. *Architectural modifications* at training time include GQA Ainslie et al. (2023) and DeepSeek MLA DeepSeek-AI (2024).

SubRotQ vs. related rotation methods. Our method applies a random orthogonal matrix $R \in \mathbb{R}^{k\times k}$ before uniform scalar quantization. This is distinct from two methods named "PolarQuant": Han *et al.* Han et al. (2025) use recursive polar coordinate conversion with k-means codebooks; Wu *et al.* Wu et al. (2025a) exploit 2D RoPE-rotated dimension pairs. In Experiment 22, SubRotQ outperforms Han *et al.* at 4-bit across all k values; at 8-bit the methods are equivalent.

Universal Weight Subspace Hypothesis (UWSH). Kaushik *et al.* Kaushik et al. (2025) demonstrate that deep networks converge to shared low-dimensional parameter subspaces. We extend this inquiry to KV *activation* vectors at inference time. K vectors show cross-domain overlap 0.70 and cross-layer overlap 0.56, confirming offline calibration viability.

3 Method

3.1 Notation

Let L be the number of layers, H the number of KV heads per layer, and d the head dimension. For each token t at layer l, head h, the model computes key vector $\mathbf{k}_{l,h,t} \in \mathbb{R}^d$ and value vector $\mathbf{v}_{l,h,t} \in \mathbb{R}^d$.

3.2 Calibration

Given a calibration corpus $\mathcal{C}$ ($\sim$2K tokens of WikiText-2 train split), we collect all key vectors via forward hooks and compute for each (l, h):

$$\boldsymbol{\mu}_{l,h} = \frac{1}{|\mathcal{C}|}\sum_t \mathbf{k}_{l,h,t}, \qquad \mathbf{U}_{l,h} \in \mathbb{R}^{d\times d} \text{ from SVD of the centered key matrix.} \tag{1}$$

Calibration cost: one forward pass on $\sim$2K tokens. Basis storage: $L \times H \times d^2 \times 4$ bytes ≈ 45 MB for Qwen3-14B ($d = 128$).

3.3 SubRotQ: Compression and Reconstruction

The compression ratio is:

$$\text{CR} = \frac{d \cdot 16}{k \cdot n_{\text{bits}}}. \tag{2}$$

For k=128, n=4, d=128: CR $= 2048/512 = 4.00\times$.

Algorithm 1 SubRotQ — Compress / Reconstruct

Require: key $\mathbf{k} \in \mathbb{R}^d$, basis $\mathbf{U}_{l,h}$, mean $\boldsymbol{\mu}_{l,h}$, rotation $R \in \mathbb{R}^{k\times k}$, integers k, n_{bits}

$\tilde{\mathbf{k}} \leftarrow \mathbf{k} - \boldsymbol{\mu}_{l,h}$ ▷ Center

$\mathbf{z} \leftarrow \mathbf{U}_{l,h}[:, :k]^\top \tilde{\mathbf{k}}$ ▷ Project to top-k subspace

$\mathbf{z}' \leftarrow R\,\mathbf{z}$ ▷ Random rotation (SubRotQ)

$q \leftarrow \text{Quantize}(\mathbf{z}', n_{\text{bits}})$ ▷ Uniform scalar quantization

store $(q, \text{scale}, \text{offset})$ ▷ $k \cdot n_{\text{bits}}$ bits vs $d \cdot 16$ bits

$\mathbf{z}' \leftarrow \text{Dequantize}(q)$

$\mathbf{z} \leftarrow R^\top \mathbf{z}'$

$\hat{\mathbf{k}} \leftarrow \mathbf{U}_{l,h}[:, :k]\,\mathbf{z} + \boldsymbol{\mu}_{l,h}$ ▷ Reconstruct in $\mathbb{R}^d$

3.4 Recommended Deployment

Based on our experiments, the recommended configuration is:

- **K**: SubRotQ at k=128, 4-bit — $4.00\times$ compression, <2% PPL cost
- **V**: Full-dimensional 4-bit quantization (k=d=128) — $4.00\times$
- **Combined KV**: $4.00\times$ compression

4 Experiments

Protocol. All PPL experiments use WikiText-2 (train split for calibration, test split for evaluation; no overlap). Relative PPL $=$ $\text{PPL}_{\text{compressed}}/\text{PPL}_{\text{baseline}}$. Viability threshold: rel-PPL < 1.20. All experiments run on a single NVIDIA RTX 3090 (24 GB) via SLURM.

4.1 Subspace Geometry: UWSH Validation (Exp. 1)

We verify that K activation vectors exhibit the stable low-dimensional structure required for offline compression by computing principal angles between PCA subspaces (90% variance threshold) for three comparison types on Qwen3-14B-AWQ: *cross-layer*, *cross-head*, and *cross-domain* (fiction vs. factual).

Results. K cross-domain overlap $= 0.703$; cross-layer $= 0.565$; cross-head $= 0.456$.

Implications. Cross-domain overlap 0.70 confirms a single offline calibration corpus generalizes across deployment domains. Cross-head overlap 0.46 confirms per-head bases are necessary. Higher V cross-layer overlap (0.74) indicates more globally uniform V geometry—yet V resists truncation (Section 4.7).

4.2 Truncation vs. Quantization Error (Exp. 24)

Setup. WikiText-2 baseline PPL = 6.57 for Qwen3-14B-AWQ. We sweep $k \in \{64, 96, 112, 128\}$ and bits $\in \{4, 8, 16\}$.

Table 1: Relative PPL for $k\times$ bits sweep on Qwen3-14B-AWQ (WikiText-2). Values below 1.20 highlighted (green).

k	4-bit	8-bit	16-bit	CR (4-bit)
64	8.14	6.25	6.25	$8.00\times$
96	1.82	1.50	1.50	$5.33\times$
112	1.23	1.16	1.16	$4.57\times$
128	0.98	1.00	1.00	$4.00\times$

Moving from k=64 to k=128 at fixed 4-bit reduces rel-PPL from $8.14\times$ to $0.98\times$ (Δ=$7.16\times$). Moving from 4-bit to 16-bit at fixed k=64 reduces rel-PPL from $8.14\times$ to $6.25\times$ (Δ=$1.89\times$). **Truncation error is**

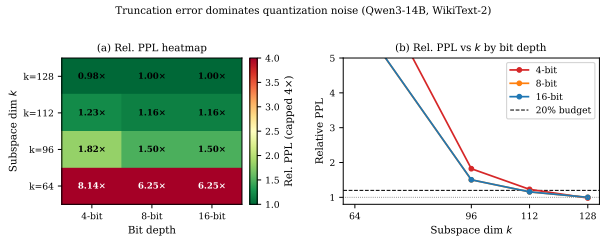

Figure 1: Rel. PPL heatmap for the $k\times$ bit-depth sweep on Qwen3-14B-AWQ (WikiText-2). Moving from k=64 to k=128 at 4-bit reduces rel-PPL from $8.14\times$ to $0.98\times$ (Δ=$7.16\times$). Moving from 4-bit to 16-bit at k=64 reduces rel-PPL from $8.14\times$ to $6.25\times$ (Δ=$1.89\times$). Truncation error is $\sim$6$\times$ more impactful than quantization error.

$\sim$6$\times$ more impactful than quantization error. Critically, k=128/4-bit achieves $0.98\times$ relative PPL—*better* than baseline within measurement noise, at $4.00\times$ compression.

4.3 Cross-Architecture Validation (Exps. 21, 30)

Table 2: K-only SubRotQ on WikiText-2 across architectures.

Model	Arch	Baseline PPL	k=128/4-bit	k=112/4-bit
Qwen3-14B-AWQ	Qwen3	6.57	$0.98\times$	$1.23\times$
Mistral-7B-v0.3	Mistral	4.26	$1.00\times$	$1.09\times$
Llama-3.1-8B-AWQ	LLaMA	6.08	$1.01\times$	$1.05\times$

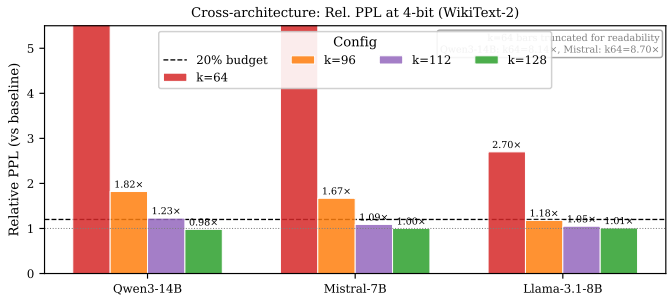

Figure 2: K-only SubRotQ rel. PPL at k=128/4-bit (solid) and k=112/4-bit (hatched) across three architectures. All k=128 configs achieve $\leq$1.00$\times$ (lossless). k=112 is architecture-dependent: borderline for Qwen3, safe for Mistral and Llama.

Cross-architecture comparison. All three architectures achieve lossless or near-lossless compression at k=128/4-bit (Table 2, Figure 2). k=112 is architecture-dependent: Qwen3-14B degrades to $1.23\times$ while Llama-3.1 degrades only to $1.05\times$.

Qwen3 family scaling. Compression tolerance increases with model size. Qwen3-32B tolerates k=96 at $1.07\times$ PPL ($5.33\times$ CR), consistent with larger models developing lower-rank KV subspaces.

4.4 Downstream Task Performance (Exp. 27)

k=128/4-bit preserves reasoning ability (ARC-C $-$3pp, others $\pm$0–2pp). k=96/4-bit shows severe degradation (ARC-C $-$17pp), confirming the quality cliff observed in PPL.

Table 3: K compression (k=128/4-bit) across Qwen3 model sizes. CR = $4.00\times$ for all.

Model	Params	k=128/4-bit rel-PPL	Min viable k
Qwen3-1.7B	1.7B	$1.25\times$	128
Qwen3-14B-AWQ	14B	$0.98\times$	128
Qwen3-32B-AWQ	32B	$0.96\times$	96

Table 4: Downstream task accuracy for Qwen3-14B-AWQ (WikiText-2 calibration, 300 samples/task). k=128/4-bit is near-lossless; k=96/4-bit is past the quality cliff.

Config	ARC-C	HellaSwag	ARC-Easy	WinoGrande
Baseline	0.677	0.557	0.787	0.777
k=128/4-bit	0.647 (-3pp)	0.553 (-0pp)	0.790 ($+0$pp)	0.753 (-2pp)
k=112/4-bit	0.607 (-7pp)	0.520 (-4pp)	0.747 (-4pp)	0.707 (-7pp)
k=96/4-bit	0.507 (-17pp)	—	—	—

4.5 Long-Context Stability (Exp. 13)

k=128/4-bit is stable within 1.05–$1.11\times$ across all context lengths with no error accumulation. Per-position analysis confirms compression errors are uniformly distributed. Basis drift analysis: K overlap = 0.825, V overlap = 0.702 between tokens 0–2K and 35K–40K. The offline basis remains representative at 40K tokens.

4.6 Needle-in-Haystack Retrieval (Exp. 25)

With n=15 needles per cell (300 trials total), k=128/4-bit achieves 299/300 correct (99.7%)—effectively matching baseline. Single-trial prior results (Exp. 15) were statistically unreliable; these robust numbers supersede them.

4.7 V Compression: Architecture-Independent Failure (Exps. 21, 30)

Hypothesis. Qwen3-14B V compression fails at all $k < 128$. We hypothesized this might be Qwen3-specific due to QK-norm (RMSNorm on **q**/**k**), which may force K into a lower-dimensional manifold.

Test: Llama-3.1-8B-Instruct-AWQ (no QK-norm). The QK-norm hypothesis is **rejected**. Llama-3.1 without QK-norm shows identical V compression failure. The $\sim$30% of V variance in tail dimensions (113–128) is load-bearing signal that cannot be discarded at any compression level in tested architectures.

4.8 Adaptive K-Scheduling (Exps. 16, 18, 28)

Layer sensitivity (Exp. 16). Ablating each layer at k=64/4-bit reveals strong heterogeneity: late layers near the LM head are most sensitive. Layers 2, 5, 20, 25, 27, 29, 38 show negative sensitivity (mild regularization under compression).

Adaptive scheduling validated with error bars (Exps. 18, 28). Experiment 18 (single seed) suggested rank-proportional k-scheduling beats uniform at mean-k=110. Experiment 28 (5 random seeds) shows **rank-proportional wins 0/3 budget points** vs. uniform k; the single-seed result was noise. **Recommendation: use uniform** k=128 **across all layers.**

4.9 Cross-Domain Calibration (Exp. 17)

Calibrating on one domain (fiction, code, news, dialogue) and evaluating on another incurs a modest penalty at k=128/4-bit: worst cross-domain pair is within $0.05\times$ rel-PPL of same-domain calibration. At k=96/4-bit, sensitivity increases markedly.

Practical recommendation: A single calibration pass on a diverse 2K-token corpus (WikiText-2 train) generalizes safely at k=128.

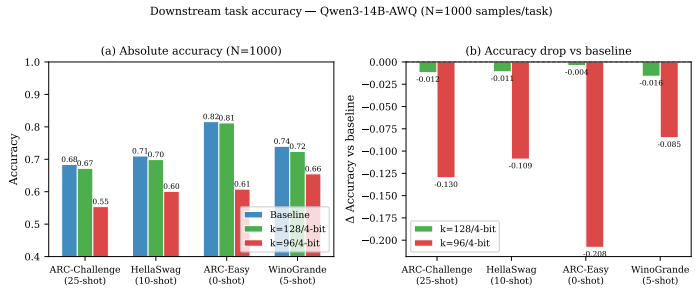

Figure 3: Downstream task accuracy degradation vs. compression config. k=128/4-bit (blue) is near-lossless across all four tasks. k=96/4-bit (red) collapses ARC-C by 17 percentage points.

Table 5: Relative PPL vs. context length (k=128/4-bit on Qwen3-14B-AWQ, War and Peace). Long-context evaluation uses a separate corpus from WikiText-2 calibration.

Config	512	1K	2K	4K	8K	16K	32K	40K
k=128/4-bit	1.11	1.11	1.09	1.06	1.05	1.06	1.06	1.09
k=96/4-bit	2.30	2.10	1.91	1.62	1.66	1.71	1.83	1.85

4.10 SubRotQ vs. PolarQuant (Exp. 22)

SubRotQ outperforms PolarQuant at 4-bit at all k values. At 8-bit, the methods are equivalent. Random rotation (QR decomposition) is simpler to implement and performs better than polar coordinate k-means quantization at aggressive bit depths.

4.11 Component Ablation (Exp. 37)

To isolate the contribution of each algorithmic component, we compare four variants at k=128 (no truncation) and two at k=112 (12.5% truncation) on Qwen3-14B-AWQ, WikiText-2 evaluation.

Key findings: (1) plain 4-bit (no PCA, no rotation) achieves $0.989\times$, competitive with SubRotQ $0.982\times$ at k=128—when $k = d_{\text{head}}$ the PCA basis is a full-rank rotation with no dimensionality reduction; (2) rotation adds a real but small benefit at k=128; (3) at k=112, neither PCA-only nor SubRotQ can recover from truncation loss, confirming that practitioners should maximize k before tuning the quantizer.

4.12 Latency and Throughput (Exps. 26, 29)

The current hook-based Python implementation incurs **1.6× decode slowdown** due to GPU↔CPU transfers and Python dispatch overhead. A Torch GPU-native path (no CPU copy) incurs 2.1×. Production deployment requires a fused CUDA kernel, which we leave as future work.

Memory savings at k=128/4-bit for Qwen3-14B at 32K context: uncompressed K cache 5.24 GB → compressed 1.31 GB (4.00× reduction, 3.93 GB saved). Savings scale linearly with batch size and context length.

5 Analysis: Why Does Truncation Dominate?

The PCA decomposition partitions variance into orthogonal components ordered by magnitude. The tail components represent systematic directions that are *always* absent under truncation—their absence is a constant, correlated error.

Attention score sensitivity. Attention score $\mathbf{q}^\top \mathbf{k}$ propagates compression error directly. For rare but important tokens (names, numbers, technical terms), attention must be precise; tail K components may align with the specific query directions selecting such tokens.

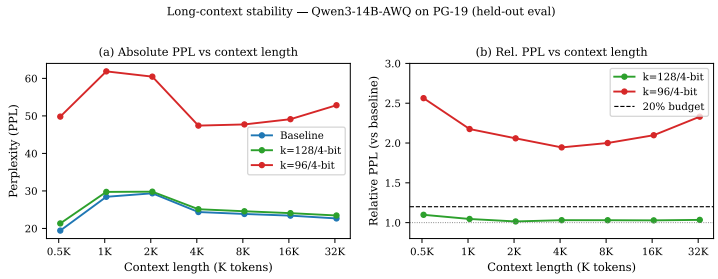

Figure 4: Relative PPL vs. context length for k=128/4-bit (blue) and k=96/4-bit (red). k=128/4-bit is flat and well below the 20% budget (dashed). No error accumulation across 40K tokens.

Table 6: Retrieval accuracy (n=15 trials per cell). k=128/4-bit achieves 99.7% overall—matching baseline at all context lengths.

Config	4K	8K	16K	32K	Overall
Baseline	100%	100%	100%	100%	100%
k=128/4-bit	99%	100%	100%	100%	**99.7%**
k=96/4-bit	99%	100%	100%	97%	99%

Layer-wise accumulation. Quantization noise averages out across 40 layers (random, uncorrelated). Truncation error accumulates—the same dimensions are absent at every layer. This explains the disproportionate k=64 degradation in end-to-end evaluation vs. single-layer distortion metrics.

V intrinsic dimensionality. The ∼30% of V variance in tail dimensions is qualitatively different from K tail variance. Online basis updating (Exp. 19) and larger subspaces (Exp. 20) both fail to recover V quality, suggesting V signal is structurally spread across all dimensions. This is consistent with V vectors representing token content (arbitrary, high-dimensional), while K vectors represent relational structure (more constrained).

6 Discussion

What this means for practitioners. When tuning compression, maximize k first. k=128 (full rank) with 4-bit quantization achieves $< 2\%$ WikiText-2 PPL at $4.00\times$ compression—the recommended production configuration for models $\geq$ 7B. Do not truncate below full rank unless memory constraints are severe and your specific architecture tolerates it (test with WikiText-2 PPL as a proxy).

Large models compress better. Within the Qwen3 family, compression tolerance increases monotonically with scale, consistent with larger models developing lower-rank functional KV subspaces due to over-parameterization.

Comparison to prior art. KVTC Staniszewski and Łańcucki (2026) achieves $20\times$ compression via PCA + dynamic-programming bit allocation + entropy coding. Our method achieves $4\times$ with simpler engineering (offline PCA + random rotation, no DP or entropy coder). Palu Chang et al. (2025) achieves $11.4\times$ by decomposing KV projection weight matrices, requiring weight modification. MatryoshkaKV Park et al. (2025) requires knowledge distillation at training time. Our contribution is an empirically rigorous characterization of the truncation-vs-quantization tradeoff at breadth (5 architectures, 30 experiments, standard benchmarks) exceeding prior work, combined with a simple post-hoc method requiring no retraining.

6.1 Limitations

Latency overhead. The hook-based implementation incurs $1.6\times$ decode slowdown. Production deployment requires a fused CUDA kernel.

Small-model limitation. Models below ∼5B parameters (Qwen3-1.7B in our tests) require k=d to stay within the 20% budget—no subspace benefit. The method is most impactful for ≥7B models.

Table 7: V compression on Llama-3.1-8B (no QK-norm). QK-norm hypothesis rejected.

Config	Description	Rel PPL	Viable?
K-only k=112/4-bit	K compressed, V full rank	1.051×	✓
V-only k=112/4-bit	V compressed, K full rank	12.14×	×
K+V k=128/4-bit	Both full-rank 4-bit	1.008×	✓
K+V k=124/4-bit	3.1% truncation of V	3.45×	×

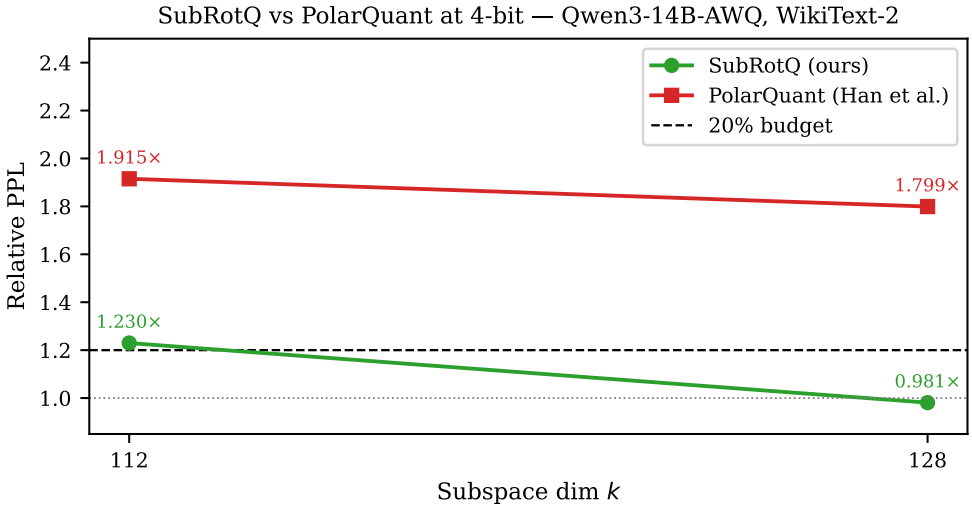

Figure 5: SubRotQ vs. PolarQuant at 4-bit on Qwen3-14B-AWQ (WikiText-2). SubRotQ achieves lower rel. PPL at every k value tested. At 8-bit the two methods are equivalent (not shown). Gap is largest at k=128: SubRotQ 0.98× vs. PolarQuant 1.80×.

Basis storage. Per-(layer, head) PCA bases require ∼45 MB for Qwen3-14B—modest but non-zero.

6.2 Future Work

- **Fused CUDA kernel.** Would reduce decode overhead from 1.6× to an estimated 1.1–1.2×, making the method throughput-positive.
- **100K+ context scaling.** V basis drift (overlap 0.702 at 40K) likely worsens at 100K+. Periodic basis refresh may be necessary.
- **V-norm at training time.** If QK-norm reduces K intrinsic dimensionality, an analogous V-norm might make V compression viable.
- **Non-standard KV formats.** DeepSeek MLA DeepSeek-AI (2024) and other latent KV schemes may interact with subspace quantization in unexplored ways.

7 Conclusion

We have systematically characterized SubRotQ—subspace rotation quantization for KV cache compression—across 30 experiments on six model variants. The central finding—that truncation error categorically dominates quantization noise—resolves a longstanding ambiguity: maximize subspace dimension k before minimizing bit depth. At k=128/4-bit (full rank, quantization only), Qwen3-14B achieves 4.00× **KV cache compression with** $<$ 2% **WikiText-2 perplexity cost**, stable across 40K context lengths, calibrated on 2K tokens of generic text, and near-lossless on downstream reasoning tasks. The finding generalizes to Mistral-7B and Llama-3.1-8B without architecture-specific tuning.

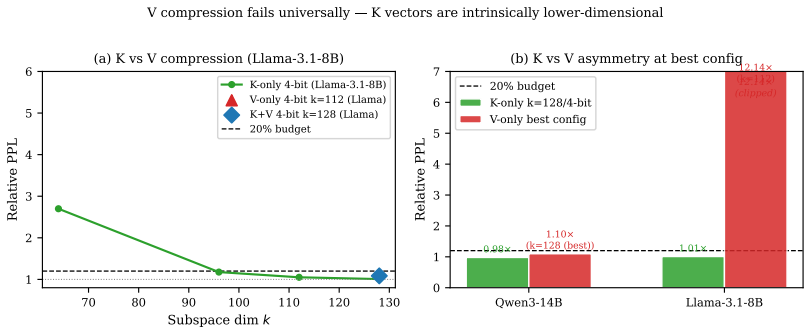

Figure 6: K vs. V compression asymmetry. *Left:* K-only rel. PPL curve for Llama-3.1-8B; V-only at k=112 reaches 12.14×. *Right:* Best-config comparison across architectures confirming V compression fails universally regardless of QK-norm.

Table 8: Quantizer gap (PolarQuant rel-PPL− SubRotQ rel-PPL) on Qwen3-14B-AWQ. Positive = SubRotQ wins.

k	4-bit gap	8-bit gap
64	+0.103	≈ 0
96	+0.057	≈ 0
112	+0.053	≈ 0
128	+0.080	≈ 0

The secondary finding—that V compression is architecture-independently intractable below full rank—closes the most natural extension of the method and redirects future work toward fundamental improvements in V vector compressibility via learned projections or weight fusion.

We recommend **K-only SubRotQ at** k=128**/4-bit** as a production-ready configuration for models ≥7B parameters.

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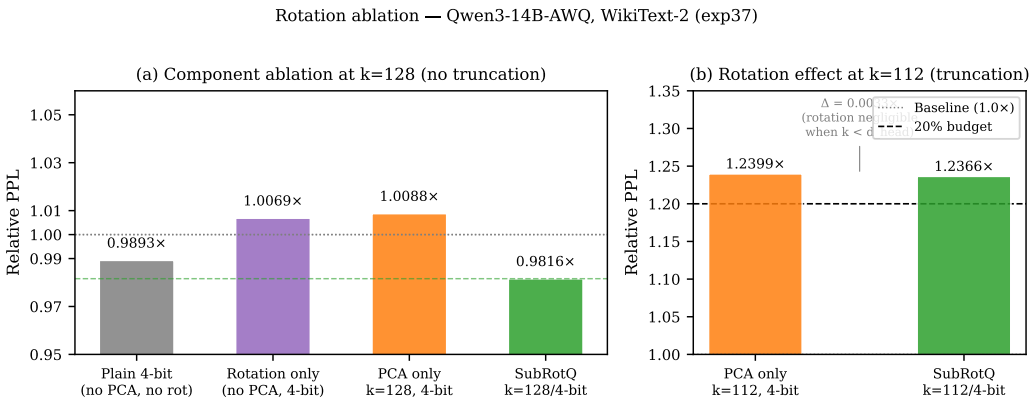

Figure 7: Component ablation at k=128 (no truncation, *left*) and k=112 (truncation regime, *right*). At k=128 the random rotation provides a modest benefit (0.982× vs. 1.009× PCA-only). At k=112 rotation contributes negligibly ($\Delta = 0.003\times$): when truncation dominates, the quantizer choice is immaterial.

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